

MENDEL UNIVERSITY IN BRNO

Faculty of Regional Development and International Studies



Master Thesis

**An Assessment of the Free Maternal Healthcare Policy on Maternal
and Infant Mortality in Krowor Municipal Assembly, Ghana**

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DECLARATION

I hereby declare that this thesis entitled *An Assessment of the Free Maternal Healthcare Policy on Maternal and Infant Mortality in Krowor Municipal Assembly, Ghana* was written and completed by me. I also declare that all the sources and information used to complete the thesis are included in the list of references. I agree that the thesis could be made public in accordance with Article 47b of Act No. 111/1998 Coll., Higher Education Institutions and on Amendments and Supplements to Some Other Acts (the Higher Education Act), and in accordance with the current Directive on the publishing of the final thesis.

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In Brno, Date.....

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DEDICATION

This thesis is dedicated to all midwives and other care providers working tirelessly to improve the infant and maternal healthcare system in Ghana. Also, to millions of humans who lost their life to infant and maternal mortality cases. May their souls rest in peace.

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My first appreciation goes to the Lord God Almighty for his grace and guidance during this study. Indeed, He makes all things beautiful in His time.

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ABSTRACT

This study sought to assess the FMHCP impact on maternal mortality and childcare. The study focused on investigating the spatial impact of the policy on maternal mortality, analyzing the policy's impact on infant mortality, examining the correlation between the FMHCP and the access to healthcare by lactating mothers and their infants and exploring the barriers to the access of this policy.

The study was underpinned by the theory of planned behavior and the health belief model. For this study, the mixed methods approach was adopted. Using a purposive sampling strategy, 96 respondents were selected to either respond to a questionnaire or interview.

From the study, it was revealed that the Free Maternal Healthcare policy has led to a decrease in maternal deaths and infant deaths in the Krowor municipality. It was further evinced that the policy has a correlation with maternal and infant access to health services, thus incrementing the usage of infant and maternal health care services. The study also revealed some barriers associated with the policy including long waiting hours at the facilities, additional costs incurred at the facilities, poor management, the attitude of health professionals, and access to the facility.

Keywords: Maternal Mortality, Infant Mortality, Healthcare Access, Facility Delivery, Free Maternal Healthcare Policy.

ABSTRAKT

Tato studie posuzuje dopad strategie bezplatné mateřské zdravotní péče (FMHCP) na mateřskou úmrtnost a péči o děti. Studie se zaměřuje na zkoumání prostorového dopadu strategie na mateřskou úmrtnost, analyzuje vliv strategie na kojeneckou úmrtnost, posuzuje souvislosti mezi FMHCP a přístupem ke zdravotní péči ze strany kojících matek a jejich kojenců a zkoumá překážky komplikující přístup k této strategii.

Studie byla založena na teorii plánovaného chování a na modelu zdravotního přesvědčení. Byla použita metoda smíšeného výzkumu. Metodou záměrného výběru bylo vybráno 96 respondentů, kteří buď vyplnili dotazník, anebo odpovídali na otázky v přímém rozhovoru. Ze studie vyplynulo, že strategie bezplatné mateřské zdravotní péče vedla ve správní jednotce Krowor k poklesu mateřské a kojenecké úmrtnosti. Dále bylo zjištěno, že strategie má vliv na přístup ke zdravotním službám ze strany matek a kojenců. Studie také odhalila jisté překážky komplikující přístup k této strategii, mj. dlouhou čekací dobu v zařízeních, zvýšení nákladů na provoz zařízení, nevyhovující management, postoj zdravotnických pracovníků a přístup do zařízení.

Klíčová slova: Mateřská úmrtnost, Kojenecká úmrtnost, Přístup ke zdravotní péči, Porod ve zdravotnickém zařízení, Strategie bezplatné mateřské zdravotní péče.

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LIST OF ABBREVIATION

AIDS	Acquired Immunodeficiency Syndrome
DHS	Demographic and Health Survey
FMHCP	Free Maternal Healthcare Policy
GHO	Global Health Observatory
GHS	Ghana Health Service
GMHS	Ghana Maternal Health Survey
GSS	Ghana Statistical Service
HIV	Human Immunodeficiency Virus
MDGs	Millennium Development Goals
NHIS	National Health Insurance Scheme
SDGs	Sustainable Development Goals
SIDS	Sudden Infants Death Syndrome
TPB	Theory of Planned Behaviour
TRA	Theory of Reasoned Action
UN	United Nations
UNICEF	United Nations Education Fund
UNPF	United Nations Population Fund
WB	World Bank
WHO	World Health Organization

CHAPTER ONE

INTRODUCTION

1.0 Chapter Overview

This chapter introduces the study, provides background to the study, and identifies the research gap which needs to be filled. It also provides the research objectives, research questions, significance and scope of the study, operational concepts, and limitations of the study. Finally, it provides a guide for the organization of the study.

1.1 Background to the Study

Irrespective of various policies set up by international communities to decree drastically the level of maternal and infant mortality, they are still of grave health concern globally. Maternal mortality has been the main cause of death in women aged 15 to 49 across the world. Over 1500 women die every day from pregnancy-related causes, resulting in a yearly total of 550 000 maternal fatalities (UN, 2009). Nearly 152 women died per 100,000 live births in 2020, with most of these deaths occurred in Sub-Saharan Africa. (Goalkeepers Report, 2021). It is also reported that in 2017, about 810 women lost their lives each day due to pregnancy and childbirth-related complications in the world. With most of these situations happened in the Sub-Sahara Africa, could have been avoided. (WHO, UNICEF, UNPF & World Bank, 2019). Globally, maternal mortality main causes have been hemorrhage, eclampsia, unsafe abortion, infections, and obstructed labor (Der *et al.*, 2013).

Belarus, Poland, Italy, and Norway had the lowest maternal mortality ratio in the world, with 2 per 100 000 live births. with the disturbingly high rate in South Sudan with 1,150 deaths per 100 000 live births in 2017. In Africa, Egypt had the least maternal mortality ratio of 37, whilst that of Chad

and Somalia documented the highest maternal mortality ratio of 1140 and 1150, respectively. (WHO, UNICEF, UNFPA and the World Bank, 2019). These figures depict a major difference in maternal health, with sub-challenge regardless of numerous mitigating strategies setup to salvage it by the global community.

The issue of infant mortality is same as the above, it is disturbingly high in sub-Sahara Africa as compared to Europe. These issues affect the sub-region greatly and has grave consequences on the development of the sub-region. It is thus not surprising that the Millennium Development Goals (MDGs) touched on the importance to reduce maternal and child mortality significantly.

The MDGs highlighted the need for maternal and child mortality to be reduced. In Ghana, maternal mortality and infant mortality is a serious menace that sometimes scares people from being pregnant and even bearing children. In 2007, the results from Ghana's Demographic and Health Survey proved that the country was nowhere near achieving the MDGs aim to reduce maternal and child mortality. To attain this goal, the Ghanaian government implemented the Free Maternal Health Care Policy (FMHCP) in 2008, which aimed to increase women healthcare access and reduce maternal and infant mortality (Johnson *et al.*, 2015). Under the FMHCP, all expectant and lactating in the country are entitled to free antenatal, delivery services and postnatal care. With the inception of this policy the incidence of maternal and infant mortality has reduced. However, it persists in Ghana, more particularly in rural and poor urban settlements. There is the urgency to reduce this public health threat to the very least for the development of the country, the sub-region, and the world at large.

1.2 Problem Statement

The issue of maternal and infant mortality is a serious challenge for the global community. Globally, sub-Saharan Africa accounts for the high index of maternal mortality (Vogel *et al.*, 2016)

and infant mortality (Kiross, 2020). There has been an uneven improvement as a result of the unequal distribution of infrastructural facilities, essential drugs, quality care and skilled human resource (Vogel *et al.*, 2016). Previous studies identified financial capacities, poor care, low expenditure on health, level of female education attainments, per capita income, general environmental cleanliness cultural beliefs and access to healthcare as the main reasons for the prominent levels of infant and maternal mortalities (Kyei-Nimako *et al.*, 2016; Arthur & Oaikhenan, 2017). Based on the ubiquitous nature of these mishaps, the United Nations in 2008 adopted the Millennium Development Goals (MDGs) with sub-goals 4 and 5 which aimed at improving child and maternal healthcare in general. Though, countries in the sub-Saharan African region have achieved marked improvement in the levels maternal and infant mortality since the inception of the MDGs, they continue to top the charts in all global regions on same issues.

Ghana, a country within the sub-Saharan region has over the years suffered from its prolonged battle with maternal and infant mortalities. The country has implemented several policies aimed curbing and reducing maternal and infant mortality. These policies include the introduction of National Health Insurance Scheme (NHIS) and the FMHCP among several others which focused on enhancing the opportunities of expectant and lactating mothers and their infants to have unhindered healthcare services access (Anarwat *et al.*, 2021). In September 2003, through the NHIS, all pregnant women were exempted from paying a delivery fee in public and religious health facilities. This was followed by the implementation of the FMHCP in July 2008. These policies were to an extent designed to increase the patronage of healthcare on antenatal, delivery and post delivery services (Adu *et al.*, 2018). These policies have paid off as there has been noteworthy improvement in maternal mortality. Though these mortalities are declining, it is at a slower rate

and resulted in the country's inability to achieve MDGs 4 and 5 like many countries in the world (UNICEF, 2019; UNCF, 2015).

The Sustainable Development Goals (SDGs), with a goal to achieve a “guarantee healthy lifestyles and promote wellbeing for all at all ages” which are targets that all countries pledged to try to meet by 2030 (Le Blanc, 2015) which include infants and pregnant women. With the FMHCP and the NHIS, Ghana was hopeful in attaining this goal. Since the inception of the FMHCP, there have been numerous studies to assess the policy. While there have been studies on variations in access to this policy (Azaare *et al.*, 2020; Anarwat *et al.*, 2021), critically missing the discussion has been spatiotemporal dimension of access outcomes. Specifically, about a decade and a half into the implementation to the FMHCP, how has the issue of maternal mortality changed over the years in Ghana? and how could these changes be explained? It is on this notion that this study is designed to fill these empirical gaps and to assess the policy on maternal and infant mortality in the Krowor Municipal Assembly in Ghana.

1.3 Research Objectives

The study sought to assess the FMHCP's impact on maternal mortality and childcare. The researcher was particularly interested in the relationship between the policy and maternal and infant mortalities. The specific research objectives were.

1. To investigate the spatial impact of the policy on maternal mortality.
2. To analyses the policy's impact on infant mortality.
3. To examine the correlation between the FMHCP and the access to healthcare by expecting and lactating mothers and their infants.
4. To explore the barriers to the access of this policy.

1.4 Research Questions

The following questions frame this study.

1. What is the impact of the FMHCP on maternal mortality?
2. How does the FMHCP impact infant mortality?
3. Is there a correlation between the FMHCP and the access to healthcare by expecting and lactating mothers and their infants?
4. What are the barriers to the access of the FMHCP?

1.5 Significance of Study

A study of this nature when concluded will provide the readers an understanding of the variables in the study and raises awareness about the various trends of the issues of maternal and child healthcare and maternal and infant mortality situation in Krowor Municipal Assembly. Significantly, the findings will be used as a reference resource by the government, health authorities, and other stakeholders in developing policies and strategizing solutions to the problem. The findings will be extremely useful in improving maternity and childcare services by strengthening institutional knowledge, national policy, and decision-making.

Although there is a plethora of anecdotal and empirical studies on the FMHCP and maternal healthcare, the added element of infants or childcare to the FMHCP to the study is a novelty and there exist a dearth of studies on the discourse. Also, there is little work done on these studies in municipal areas, with most often such studies are done in the rural areas. As a result, this study will add to the body of information on the topic, allowing for subsequent research by other researchers to widen the scope or act as a baseline for future studies and a framework for monitoring changes in these issues.

1.6 Scope of the Study

The research scope delimits the study. The research scope according to Padua & Santos (2008) is determined by study population and sample and variables used in the study. This study is confined to the Krowor Municipality in the Greater Accra region.

1.7 Limitation of the Study

Every study has its limitations, and it would be erroneous if those factors are not pointed out. The busy nature of the municipal health officials and bureaucratic processes was a major problem. Occasionally scheduled meetings were canceled and, I was to seek permission from higher authorities at regional level before they could give me access and do interview with me. Also, an official I interviewed instructed me not to write or record what he/she was saying. There exist a limited literature and academic work on the topic in the study area and few existing literatures focused on other geographical locations.

The above limitations, however, do not render the findings of this research unreliable and inapplicable since the researcher managed these limitations effectively to ensure that the research objectives of the study were achieved.

1.8 Operational Concepts

The operational definitions of important terminologies utilized in the study are presented in this section. They are discussed in a way that communicates the study's actual meaning.

Maternal healthcare: Antenatal care, delivery and post-natal services provided to expectant and lactating mothers.

Antenatal care: This refers to visit before delivery period and normal number of antenatal care visit is usually four times in Ghana.

Facility delivery: It is the period of childbirth by expectant mothers or pregnant women in an approved health facility either by self or through surgery.

Maternal mortality: Deaths occurring in women because of pregnancy-related and childbirths complications.

Infant mortality: This is when a baby dies before reaching his or her first birthday.

1.9 Organization of the study

The report is organized into five chapters with each chapter dealing with a particular aspect of the study. Chapter one introduces the study. Chapter two summarizes the views and thoughts of authorities who have distinguished themselves in the field of sculpture as documented in books, journals, and other literary sources. The methodology aspect of the study takes Chapter Three, which focuses on the research design, the study areas, the target population, sampling methods employed, the data collection instruments used and the sources of data for the study. It also includes ethical issues considered in data collection and the data analysis.

The fourth chapter presents the data analysis and the discussion of issues relating to it. It also interprets the data and its findings corresponding to the objectives set out for the study. Chapter five summarizes and conclude the study and offer recommendations for policy and future studies.

CHAPTER TWO

LITERATURE REVIEW

2.0 Chapter Overview

The chapter dealt with the review of related literature on the discourse. It takes into consideration theories and some of the relevant areas such as the maternal health, infant healthcare, FMHCP, maternal and infant mortality. The review of literature is based on its utility to this study.

2.1. Theoretical Underpinning

A review on articles involving health behaviors theories found the Theory of Planned Behavior and Health Belief Model as the most often used theories and would be employed in this study

2.1.1The Theory of Planned Behaviour

Planned Behavior Theory (TPB) was developed by Ajzen (1991) as an extension of Theory of Reasoned Action (TRA) developed by Fishbein and Ajzen (1975). The fundamental conjecture of TRA is that people are rational beings who make rigorous considerations of the consequences of engaging or not engaging in prescribed behavior based on the information accessible to them. (Ajzen & Fishbein, 1980). One's health behavior attitude is their subjective view of how good or bad they will feel if they engage in a specific health behavior. The subjective norms refer to the individual's take on how their significant others feel about their decision or otherwise to engage in a said behaviour (Fishbein & Azjen, 1980). These significant others are people whose viewpoint on the individuals matter to the latter (Conner & Norman, 2005). The TPB adds the factor of perceived behavioral control to the assumptions of the TRA, that is an important determinant of health behaviors. In the perceptive view of Conner and Norman (2005), perceived behavioral

control is the degree of an individual's conception of his or her control over person and extraneous factors which may either facilitate or constrain the performance of a targeted behaviour. It is a representation of a person's belief in the availability and convenience of all necessary resources needed to perform such behaviour (Conner & Norman, 2005).

A person may have the intent to adopt a new health behaviour however, factors perceived to be within or beyond his or her control may influence the targeted behaviour. For example, a pregnant woman who wishes to attend maternal healthcare at approved health facilities but issues of cost, distance, access, socio-cultural norms may affect her carrying out the targeted behaviour.

2.1.2 The Health Belief Model

This model is a cognitive based one and emanated from the psychological model developed by American psychologists (Skinner, Tiro, & Champion, 2015). The theory was first developed to explain people's non-engagement in health-related programs aimed at preventing and detecting diseases even if those facilities free and are within their reach in the 1950s. The model emphasizes that the possibility of an individual to take measures to prevent their actions is hinged on individual perception of the present condition he or she is in; the perception of the effectiveness of outcome of the condition and that the gains from reducing the said threat will exceed its cost (Glanz *et al.*, 2008).

Thus, an individual's responds or otherwise of endorsed health action is hinged upon his or her assessment of available options, the profits and usefulness of the required health behaviour and related cost and abilities to perform the required health behaviour (Conner & Norman, 2005). The model postulates that educating about their health practices are of a necessity and that of the changes that can be noticed if there are any of such (Glanz *et al.* 2008). Health belief model entails five key constructs which include the following below.

Perceived susceptibility, which is an individual's subjective evaluation of their chances of getting a health problem. That is, the likelihood of an individual adopting a healthy behaviour or seeking medical care is dependent on his or her susceptibility of a particular negative health outcome (As such, there is a less likelihood to prevent a disease which has little likelihood to afflict him or her (Carpenter, 2010).

Perceived severity is an individual's perception of the level of the health problem's seriousness. The negative impacts may in Carpenter 2010, view include death, mental or physical impairment, and pain. Thus, if the outcome of a perceived health problem is not significantly negative, then the individual is less likely to act on it.

Perceived benefit refers to a person's view on how important it is to adopt new health behaviour. If an individual believes that the proposed health behavior is beneficial and will certainly avert a negative health consequence, he or she will adopt it. (Carpenter, 2010).

Perceived barriers refer to an individual's perception on the barriers adopting new health behaviour. These barriers can for the aim this study include cost, time, social norms, views of other significant people etc. People are likely to engage in a new health behaviour if they believe they have the capacity to surmount the challenges associated with it.

Cues to action are the extra factors that encourage people to change their health behaviour. These cues assist in making the decision to engage in a recommended health behavior as well as the readiness to do so (Rosenstock, 1966). These cues include but not limited to mass media advertisements, advice from physicians or relevant other, reminder postcard and internal cues like negative body changes or pains (Janz & Becker, 1984). Edberg (2007) opined that human behavior can be influenced by combination of one or any of the above-mentioned constructs.

The model helps in understanding how pregnant women perceive pregnancy and what causes them to take the actions they do. From this model, the various constructs affect the decision of a woman to make decisions on antenatal, delivery and postpartum outcomes, thereby making the women active participants in their health outcomes and decisions.

2.2 Review of Empirical Studies

Existing studies on maternal mortality seeks to define or explain the concept and find causes of the phenomenon. The subsequent paragraphs are devoted to empirical discussions on the discourse.

2.2.1 Maternal Mortality

Maternal death has been a grave issue of the entire world. It reduces a country's life expectancy and threatens the human resource of a country and consequently its productivity. Global maternal death was reduced by almost half between 1990 and 2010 (GHO, 2015). However, the incidence is still high in developing and under developing countries and most especially the rural areas. These countries most especially those in Africa has a high fertility rate of women at reproductive years and this predisposes them to elevated level of maternal mortality. In these countries where maternal healthcare is weak, poor, and lacking coupled with higher fertility rates, the resultant is higher risks of maternal death due to childbirth or pregnancy related complications. In Sub-Saharan Africa, a women have 1 in 6 probabilities of dying during childbirth, compared to 1 in 8,700 in other countries in the Global North. (WHO, UNICEF and UNFPA, 2010). About half of maternal mortality incidence were because of socio-economic factors. Thus, geographic, and socio-economic factors influence maternal mortality (Bishai *et al.* 2016; Tlou *et al.*, 2017).

2.3 Causes of Maternal Mortality

Studies is replete with numerous findings on the causes of maternal mortality (Adu-Bonsaffoh *et al.*, 2013, Ameyaw *et al.*, 2016; Budu, 2020, Der *et al.*, 2003). These causes are categorized in medical, socio-cultural factors and infrastructural and assess factors.

2.3.1 Medical Factors

Studies show that once expecting mothers are not frequent to hospitals for antenatal clinics (ANC), they expose themselves and their unborn children to a lot of pregnancy related complications and conditions which can result in their death either before, during or post-partum (Abor *et al.*, 2011; Asamoah *et al.*, 2011; Awoonor-Williams, 2013; McCaw-Binns, 2004). These medical causes are categorized into direct and indirect (WHO/UNICEF, 2007).

2.3.2 Direct medical causes

Abou Zahr (1995) identified these direct causes are those causes that occur only during or immediately after the post-delivery period. According to research conducted by Say *et al.*, 2014, these causes account for about 73% of global maternal deaths. These causes are preventable provided there is proper healthcare and timely access to skilled healthcare, but in most developing countries these direct causes are the main factors contributing to maternal deaths (WHO, 2015). These complications include.

a. Unsafe Abortion

Unsafe abortion is undeniably a major cause of death of many women within the reproductive age. A study in Ghana reveals that a one out of every four pregnancies from adolescent females (15-19

years) result in induces or spontaneous abortions (GSS and Macro International, 1998). In Ghana, there is stigmatization attached to unmarried women who get pregnant or give birth (Hill *et al.*, 2009) and these women may tend to go in for abortions. Severe bleeding, lower abdominal aches, and the passage of fetal and placental materials are all signs of an abortion. Other studies also reveal that abortion as a leading direct cause of maternal mortality. Globally, Blencowe *et al.* (2012) identified that over 42 million females decide to carry out abortion in cases of unplanned pregnancies and unsafe abortions causes at least 60, 000 maternal deaths annually. Asamoah *et al.* (2011) also identified 13.7% of maternal death being caused by unsafe abortions.

b. Obstetric Hemorrhage

Worldwide, 8 to 13. 6 million maternal mothers suffer with post-partum hemorrhage (PPH), which is characterized by severe bleeding during delivery. PPH is one of the main factors contributing to maternal mortality, accounting most of the deaths in underdeveloped countries as compared to one percent in the developed countries. (Blencowe *et al.*, 2016; Say *et al.*, 2014. Several studies have revealed hemorrhage as one of the leading causes of maternal mortality including those done in Ghana (Blencowe *et al.*, 2016; Der *et al.* 2013; Mensah *et al.*, 2011; Asamoah *et al.*, 2011).

c. Sepsis

Puerperal Sepsis, also known as Sepsis, describes infections and systematic symptoms of infections of the genital tract that begin with an obstructed rupture of membranes and last for 42 days after delivery. It frequently happens when the amniotic sac rips before delivery, when vaginal examinations are as frequent, or when labor is impeded. WHO (2012) reveal that post-partum

sepsis is less likely in instances women delivers at a good and very hygienic health facility and where early stage of infections are diagnosed and treated on time to prevent maternal death.

According to Say *et al.* (2014) 10.7% of all maternal death are associated with sepsis. The lack of necessities such as clean water, safe delivery rooms, and limited antibiotic access predisposes pregnant women in the sub-region to sepsis. In Ghana, the 2007 Ghana Maternal Health Survey identified sepsis as one of the main causes of maternal death. This finding was also corroborated in the studies conducted by Mills *et al.*, (2008), Zakariah *et al.* (2009) and Asamoah *et al.* (2011).

d. Eclampsia

This is one of the obstetric emergencies that are more challenging to prevent or manage and a leading cause of maternal mortality worldwide. Pre-eclampsia when left untreated results in eclampsia and can be fatal to a pregnant woman (WHO, 2015). Say *et al.* (2014) asserts that it accounts for 14% of all maternal death cases. It is a life-threatening disorder caused by a sudden, fast rise in blood pressure, edema, and albuminuria that occurs during pregnancy. It usually results from the presence of protein in pedal edema and urine (Singh *et al.*, 2015).

Hypertensive disorders normally progress from pre-eclampsia to eclampsia with defined conditions of severe renal failure, intracerebral hemorrhage, and pulmonary edema. If this illness is not treated on time, it might cause convulsions and brain damage, which can lead to death. In Ghana, Adu-Bonsaffoh *et al.* (2013) in a retroactive descriptive study, found that eclampsia was responsible for 63 of 199 maternal deaths between 2010 and 2011, according to retrospective descriptive research conducted at the Korle-Bu Teaching Hospital in Accra.

e. Obstructed Labour

Obstructed labor occurs when the fetus is unable to descend via the birth canal due to the presence of an impassable barrier (obstruction) that prevents it from doing so regardless of severe uterine contractions. (Sidney *et al.*, 2002). Sidney *et al.* (2002) avow that this is mostly caused by cephalo-pelvic disproportion- a mismatch between the fetal head and the mother's pelvic brim and abnormality in fetal presentation. Obstructed labor impairs the smooth descent of the baby and in extreme situations it can result is fistula, where urine and fecal matter have access to the reproductive system. According to Neilson *et al.* (2003), the number of maternal deaths because of obstructed labor or labor dystocia varies between 4% and 7% of all maternal deaths in Ghana.

2.3.1.2 Indirect medical causes

There are also the indirect causes of maternal mortality as revealed by existing studies (Zakariah *et al.*, 2009; McDonald, 2013). These conditions may be pre-existing conditions such as infectious or non-infectious diseases and other miscellaneous conditions. According to MacDonald (2013) these indirect causes make up 20% of all maternal deaths worldwide. The study evinced that, 70% of all maternal deaths were because of direct pregnancy causes while those indirect causes made up slightly over 27% of the total deaths. However, in their study, Zakariah *et al.* (2009) found that only 8% of maternal death resulted from indirect causes. The WHO (2005) list some of these indirect medical causes and they comprise; anemia, malaria, diabetes mellitus, tuberculosis, cardiac diseases, hepatitis, and HIV/AIDS.

The indirect causes, however, are due to other non-obstetric medical conditions such as malaria, HIV/AIDS, bronchopneumonia, meningitis, tuberculosis, and anemia. According to the Ghana

Statistical Service (2009), these indirect causes mostly include malaria, HIV/AIDS, hepatitis, respiratory infections, anemia, sickle cell disease, meningitis, cerebrovascular diseases, and others.

These findings corroborate that of the Ghana Demographic Health Survey which revealed that between 2003 and 2008, the number of pregnant women with anemia rose the 65% to 70%. The study also revealed that malaria was the principal cause of anemia in Ghana (GMHS, 2017).

2.3.2 Other causes

Studies reveal that several other factors have been implicated as major contributory factors to maternal death. Cultural, religious, socio-economic, poor access of health facilities, care providers, logistics, substance abuse and many have been blamed as causes for high maternal mortality in Ghana and other part of sub-Sahara Africa (GHS,2009, Abor *et al.*, 2011).

a. Substance Abuse

According to WHO (2015), alcohol consumption and smoking behaviors have effect on both the mother and the unborn. Expectant mothers who engage in alcoholism risk higher maternal death and even endanger the lives of their babies which can result in higher risk of infant mortality. The abuse of substances is a known leading cause of hypertension, anemia, diabetes, childbirth complications (WHO, 2015b). In addition, Van Lerberghe *et al.* (2014) reveal that females who abuse substance suffer from micronutrient deficiency, and this is further worsened by pregnancy and can result in mortality. A study carried out in Ghana by Asamoah and Agardh (2012) showed a positive relationship in maternal alcoholism and maternal death.

b. Cultural factors

Research carried out in developing countries with substantial risk of maternal mortality revealed that aspects of culture are some causes of maternal mortality (Gazali *et al.*, 2012) and Ganle *et al.* (2015) suggest that the cultural and religious practices and beliefs of the African have a connection with the high incidence of maternal deaths in the sub-region. They opine that some of the traditional practices in the sub-region leads to illiteracy among women in their reproductive years and affect their ability to make informed decision during pregnancies, childbirth and after birth. In addition, they claim that childlessness which leads to stigmatization in many African societies can force women to have children they are not ready for.

Traditional practices and expectations of females have been identified in the literature as a prime cause of maternal mortality (Ganle *et al.*, 2015). In the African society, women are anticipated to give birth, take care and at times to provide for the family and the children. This expectation according to Ganle *et al.* (2015) makes it a challenge for them to be educated on required standards. Most of them are betrothed as early as seven years and moved to their husband's homes. However, since these girls are not well developed to withstand the challenges of pregnancy and childbirth, most of them die during labor, especially when there is no timely medical intervention.

Another cause is taboo. In some African societies, pregnant women are prohibited from doing and taking certain foods (Darko, 1992) which affects their health conditions and lead to pregnancy related conditions and even deaths. The deeply rooted incidence of patriarchy has significant effect on the woman's reproductive health decisions. Some cultures restrict the woman in deciding matters relating to her health and leave it for the men to decide. Mothers are left to their fate during pregnancy especially in case of unwanted pregnancies.

Religious beliefs have also been identified to have a connection with maternal mortality among the pregnant women (Ganle *et al.*, 2015). These beliefs are evinced in the prenatal period. Gazali *et al.* (2012) noticed that pregnant women in Muslim homes replace antenatal clinics with Islamic rituals by leaders and reading of Qur'an. Other pregnant women also seek prayers from pastors, prophets, mallams, imams, and witchdoctors in cases of complications where medical attention is urgently needed (Ganle *et al.*, 2015). The visit to these people is to seek spiritual protection from bewitchment, protection of their life and their unborn babies and miscarriages.

In addition, some religions do not subscribe to some life-saving procedures in modern medicine. Based on religious grounds, some women refuse medical treatment and resort to unacceptable concoctions for protection. Oppong –Darko *et al.* (2017) also revealed that most religions frown on premarital sex, pregnancy without being a married person, and abortions. Hence such members tend to engage unsafe abortions to avert the punishment prescribed per their faith.

c. Socio-economic factors

Issues of poverty, age, education, access to social amenities, place of residence among other have also been identified in the literature as causes of maternal death. On age, a study by Kondan *et al.* (2017) revealed that most maternal death occur within the age groups of 20-35 years given that most births also occur within this age group.

Poverty has also been found to be a cause of maternal mortality (Abor *et al.*, 2011). Poverty prevents people particularly pregnant women from seeking medical health even if these facilities are close to them. People without steady income are unable to cater for themselves and their families. The level of income of a family determines their purchasing power and access to services. Maternal healthcare requires enough money for facility visits, buy a variety of foods to ensure that

they get adequate nutrition among others and if they are unavailable may result in maternal death (Singh *et al.*, 2015).

Closely related to poverty is education. According to Singh *et al.* (2015), mothers with low education levels tend to have an inadequate income. Also, educated people tend to have access to inform to promote their health choices and outcomes. Most mothers with high education level have enough knowledge about their health conditions that help in finding timely remedies and the use of modern contraceptives.

d. Infrastructural causes

Studies also show that the lack of adequate health facilities and logistics cause maternal mortality. The inadequacy of facilities coupled with poor conditions of transport system and inadequate economically viable opportunities have resulted to the high maternal mortality ratio (Abor *et al.*, 2011). High incidence of maternal mortality caused by poor access to healthcare services. With limited healthcare facilities and logistics especially in rural communities, results in higher maternal mortalities especially in rural communities.

2.4 Infant Mortality

Infant mortality, conceptualized as the risk of a live born child dying before its first birthday is one of the most sensitive and frequently used measure of the socio-economic development of a population.). Thus, it is a development indicator for health and socioeconomic status, quality of life and life expectancy of a population. The measures of infant mortality are beneficial in making significant contributions to the appreciation of population growth and health related programs (GSS/GHS/ICF, 2014).

Infant mortality is a global health concern globally. It threatens the human resource of a country and consequently its productivity. Infant mortality rate indicates a measure of a country's health policy, systems and practices, an aspect of its national development. According to Okwuwa and Adejo (2020), it is mostly connected with socio-economic factors of unemployment, paucity, income difference, among others, in a community. They also assert that infant mortality revolves around primary care and primary healthcare insufficiencies, and are paucity, illiteracy, and unemployment driven, normally centering on low maternal welfare. The infant mortality rate in Africa is however still high at 59 deaths per 1000 live births. Sub-Saharan Africa has the highest infant deaths in the region with a rate of 64 deaths per 1000 live births (PRB, 2015).

Infant mortality has become more prevalent due to lack of access to health care before, during and after delivery. This contributes to high infant mortality rates both in developing and under-developed country (WHO, 2017) particularly in rural places. This assertion is in line with data from the Ghana Statistical Services' demographic and health surveys over the years. Most developing countries could not achieve the MDGs goal on infant mortality despite policies and measures put in place and Ghana is no exception. This has been attributed to inadequate national data to provide complete and reliable information on child health, lack of well-motivated personnel with adequate skills in the health sector as well as socio-economic and socio-cultural factors (UNDP, 2015).

2.5 Causes of Infant mortality

Infant mortality emanates from several causes of which are, socio-cultural, socio-economic, living conditions, and medical causes. Some medical causes known to cause death in infants are low birth weight, malaria, diarrhea, malnutrition, anemia, sudden infant death syndrome and many others.

Intricately linked to low birth weights are malnutrition, anemia, and absence of exclusive breastfeeding. It is asserted that maternal under nutrition contributes to 800,000 neonatal deaths yearly through gestational age at births around the world (Obiang-Obounou & Fuh, 2020). According to Promise Renewed Progress Report 2012, 165 million children under 5 are stunted (low height for age) in their growth due to poor nutrition during the first 1,000 days of life. Stunting rates in sub-Saharan Africa have decreased from 47% in 1990 to 40% in 2011, yet the prevalence is still high. In the view of Obiang-Obounou and Fuh (2020) malnutrition in Chad is linked to problems of anaemia, absence of exclusive breastfeeding, and child wasting (i.e., a lower-than-average weight for one's height). Malnutrition in expectant mothers will make the anaemic which affect the infant as his weight at birth will be low. Low birth weight thus has an impact on infant mortality.

Sudden Infant Death Syndrome and infant mortality is another medical condition that causes infant mortality. (SIDS) is unexplained death, usually during sleep, of a healthy baby. SIDS cause remains unknown; however, doctors speculate on risk factors like babies sleeping on their stomach, exposure to cigarette smoke in the womb or after birth, premature delivery, being a twin or triplet, being born to a teen mother, and living in poverty are all factors. (Hoyert, 2012). SIDS occurs between 2 and 4 months and mostly in winter. SIDS was the third leading cause of infant deaths in United States in 2015. Other medical conditions known to cause death in infants are low vitamin A, malaria, acute respiratory infections, and diarrhoea.

Lozano et al. (2012) point out that an estimated 250,000-500,000 children in developing countries become blind each year owing to vitamin A deficiency, with the highest prevalence in Southeast Asia and Africa. Malaria is a deadly disease which is predominant in the tropical areas of the world. Malaria is part of the childhood killer diseases and several studies have pointed that malaria

also do cause death in infants. According to Promise Renewed Progress Report 2012: In 2011, malaria accounted for a loss of about 500,000 lives of children under-five in the world with all the deaths occurring in sub-Saharan Africa. On Acute respiratory infections, Onuora Okwuwa and Fuh (2020) states that “Acute respiratory infections include upper and lower respiratory tract infections commonly manifesting cough, fever, and rapid breathing. Infant mortality rates from 2009-2017 (deaths per 1000 live births) were 83.9 (2009), 81 (2010), 78.3 (2011) and 75.7 (2017)”. Acute respiratory infections are the fourth main cause of under-five mortality. World Bank (2010) postulates that Nigeria lost 41 healthy years of life per 1,000, due to acute respiratory infections. Another disease that can cause death in infants is diarrhea. It is the second most common cause of infant mortality. This position is affirmed by A Promise Renewed Progress Report 2012: their report states that diarrhea causes about 11% of under-five deaths worldwide with nine-tenths of these deaths occurring in Sub-Saharan Africa. The World Bank (2010) discloses that Nigeria lost 43 healthy years of life per 1,000 from diarrhea illnesses.

Other environmental factors that are known to threaten the life of an infant are the living conditions of the expectant mother, housing, and toilet facilities available, access to potable drinking water, source of drinking water, and place of delivery (Wak, 2002). According to an unpublished study by Wak, (2002) in the Kassena-Nankana District in Ghana, infant mortality is experienced by children whose compounds have no toilet facilities while those who use water closet or pan toilet latrine experience the lowest infant deaths. This study also indicated that children born to mothers who drink water is from unprotected sources, experience high infancy deaths than children born by mothers whose drinking water is from protected sources. Place of resident of the expectant mother being it urban or rural have been linked to risk of infant mortality with infant’s birth in rural areas being highly exposed to mortality (Adewuyi, 2017; Gyamfi ,2002). A study by

Adewuyi (2017), concluded that infants in rural areas had higher rates of mortality than their urban equivalents and inequalities in risk factors between them.

A study Gyimah (2004) on maternal education and infant mortality in Zambia revealed that, the mortality risk of infants in the rural areas is higher than in the urban areas. The results showed that mothers who live in urban centers are less likely to record infant deaths by a factor of 0.11 compared to those living in rural settings. Tettey et al (2013) studies in Ghana and Nigeria showed a major link between place of residence and infant deaths.

On the socio-economic factors of infant mortality, several studies have brought to bear that the age, level of education, employment status and social status of the expectant mother has an impact on infant mortality. Another factor is education for women (Fischetti, 2011). In all countries globally, Fischetti, (2011) asserts that whether females increase formal education from 10 years to 11 years or two years to three, infant mortality reduces. He further postulates that women with high education tend to have smaller families, due to increased employment opportunities and better knowledge about contraception; fewer children in a family improves the chances that an infant will live (Fischetti, 2011). Formal education also enables women to make better decisions about many health and disease factors such as prenatal care, basic hygiene, nutrition, and immunization- which are important to decreasing the main cause of death in children (Fischetti, 2011).

Again, a study conducted by Ntimba and Mbago, (2005), in Tanzania, found that employed mothers were about 2.3 times less likely to experience child deaths as compared to mothers who were unemployed. Others socio-economic factors that have been linked to infant mortality are the sex of the infant (WHO, 2003) opines that male infant are at higher risk of death based on biological factors. Ethnicity has also been linked to infant mortality by some studies. A study

carried out by Mustafa and Odimegwu, (2008) in Kenya point to ethnicity as one of the vital determinants of infant mortality in both urban and rural settings.

2.7.1 The FMHCP and Maternal Mortality

The FMHCP was introduced to curb the high incidence of maternal mortality (Azaare *et al.*, 2020; Dalinjong *et al.*, 2018). A study by Baffour-Awuah *et al.* (2015) have all documented some benefits- the policy has to an extent reduced maternal mortality. In the same vein, GSS (2015) and WHO (2015) corroborate that the FMHCP have significantly reduced MMR more than two-fold between 1990 and 2015 and have reduced illness and death in mothers (WHO, 2019). The FMHCP's successful implementation according to Mardieh *et al.* (2019) comes with several benefits which includes a marked improvement in maternal deaths.

Though these findings are in line with the goal of the policy, Haruna *et al.* (2019) however vocalizes that it is contentious to posit that the policy alone has reduced maternal mortality base on the fact that these mortalities are multifaceted.

2.7.2 The FMHCP and Infant Mortality

Studies reveal that there have been improved outcomes in infant mortality because of the FMHCP. For example, Dwomoh *et al.* (2020) found that in Ghana and Burkina Faso where the FMHCP policy exist, there is a marked reduction of infant mortality compared to Nigeria and Zambia where there was no such policy. They opine that the usage of the FMHCP has help in reduction of stillbirths and neonatal death. Further, WHO (2019) observed that there has been a reduction in illness and death and their newborn babies and infants, emanating from delivery care access. Azaare *et al.* (2020) perceptively avow that the FMHCP has contributed positively to the reduction in stillbirth, neonatal and perinatal deaths.

2.8 The FMHCP, Maternal and Child Healthcare

There is a dearth of studies on the relationship between the FMHCP, maternal and child healthcare. The existing literature is silent on the correlation between these constructs but vocalizes improvement in maternal and child healthcare because of the policy. For instance, Nsiah-Boateng (2015) opines that the policy introduction has led to improvements in maternal and child health care service utilisation. Similarly, in Ghana Azaare *et al.* (2020) reveal that the free maternal services are associated with increased usage of maternal and infants' healthcare services.

Below are images of healthcare services accessed by mothers and pregnant women under the FMHCP:

- Figure 1a, shows an expecting mother and two females care providers. The expecting mother is taken her monthly sulfadoxine and pyrimethamine. These two drugs are medicine given to pregnant women to prevent malaria sickness during pregnancy in Ghana.
- Figure 1b shows, expectant mothers at maternity hall waiting for delivery of new babies.
- Figure 1c, shows lactating mothers at post-natal clinic accessing health under the FMHCP.



Figure 1a: Pregnant woman at antenatal clinic with care providers



Figure 1b: Healthcare service access at maternity hall under the FMHCP



Figure 1c: Healthcare Access under the FMHCP by Lactating mothers at post-natal clinic

Source: Enyan-Mensah, 2022

2.9 Barriers to access of the FMHCP

The review of literature showed a plethora of studies on the various barriers to maternal healthcare whether free or not. Over the years, the governments have attempted to improve access to health services through some policies. However, studies found out that the policy comes with other attendant barriers which limit access for some women. The existing literature revealed some barriers to include, financial, physical, organization/institutional, and cultural. Studies by Nisingizwe *et al.* (2020), Anafi *et al.* (2018), Bintabara *et al.* (2018) and Hox (2010) have shown that about half women report at least one of these forms of barriers in accessing maternal healthcare.

Financial barriers: This is a predominant barrier which according to Shartzer *et al.* (2015) deprive many pregnant women in sub-Saharan Africa from accessing maternal healthcare. Maternal

healthcare fees vary among countries based on national policies. In Ghana, Mozambique and South Sudan women complained about other excessive cost associated with institutional delivery although is free (Kyei-Nimakoh *et al.*, 2017; Munguambe, 2016; Wilunda *et al.*, 2016). A study in Ghana by Anafi *et al.* (2018) reported that for most pregnant women, the cost of maternal healthcare irrespective of its free components comes with some direct and indirect costs. These costs involve the payment for the provision of certain antenatal services and the purchase of certain items demanded by care providers for childbirth in these approved delivery facilities, even in public health facilities. These unofficial medical charges, bills, and fees irrespective of the free policy in the perceptive view of Witter *et al.* (2013), and Agyepong *et al.*, (2016) results from inadequate funds received by public health facilities for their management and operations from government. Even though these expenses may seem minor, had an impact on maternal behavior of the women seeking healthcare in their study. In Togo, Arnold *et al.* (2016) asserts that there is a high financial cost in the case of facility-based delivery and that those who delivered in health facilities risked staying in prolonged debt. Another impediment to antenatal care visits was the cost of travelling these healthcare facilities (Kaba *et al.*, 2017; Lawry *et al.*, 2017). These financial constraints hinder women from making use of maternal healthcare whether free or otherwise.

Physical constraints: Factors such as transportation, distance, topography, and weather are also evinced to be a significant obstacle to maternal healthcare access. With transportation-related barriers being seen as a major outcome in their studies (Arnold *et al.*, 2016; Munguambe, 2016; Wilunda *et al.*, 2016). With transportation one can talk of unreliable road networks (Lawry *et al.*, 2017; Kaba *et al.*, 2017) and weather complications (Arnold *et al.*, 2016) which makes road impassable.

The issue of distance between the place of residence and health facility has also been extensively looked at in the literature (Azuh *et al.*, 2015; Mselle & Kohi, 2016). Evidence from the study of Ganle *et al.* (2014) and Wilunda *et al.* (2017) from Ghana and South Sudan suggest that women found it a challenge to leave their houses and household obligations to travel considerable distances to seek for prenatal care, delivery service, or postnatal healthcare. Arnold *et al.*, 2016 also assert that unreliable ambulance service is another health system barrier to accessing the free healthcare for pregnant and lactating mothers. In Ghana, ambulance services are also unreliable and in the case of emergency it might be fatal.

Cultural constraints: The utilization of maternal healthcare services was impacted by customs, traditions, and cultural beliefs and practices. For instance, research reveal that in Ethiopia and South Sudan, women prefer to give birth in their homes rather than health facility delivery (Lawry *et al.*, 2017; Wilunda *et al.*, 2016). Among the South Sudanese and Mozambican women, they perceived that facility-based deliveries are for complicated pregnancies. Similarly, Ethiopian women preference to deliver new babies at home is to enable them to participate in post-partum religious observation “onur” which allows a religious leader to bless a newborn immediately it is born. In certain cultures, Kohi *et al.* (2018) asserts that a woman’s decision to give birth in the hospital is dependent upon her spouse’s approval. According to the literature, husbands can compel their wives to give birth at home to avoid their wife being exposed in front of male health care providers during delivery (Kohi *et al.*, 2018).

Institutional constraints: This constraint has to do with the challenges of the health system which ranges from lack of health facilities, inadequate skilled personnel, limited or lack of machineries and equipment, long queues and waiting hours and poor-quality care (Galaa & Dandabo, 2015; Dalinjong *et al.*, 2018; Ganle *et al.*, 2014). These issues are mostly endemic in the rural areas where

maternal and infant mortality is high (Tettey, 2003; Gyamfi, 2002; Gyimah, 2004). It is therefore not surprising that studies point to the fact that maternal and infant mortality is still high in rural areas despite the introduction of the FMHCP (Galaa & Dandebo, 2015; Dalinjong *et al*, 2018; Ganle *et al*, 2014). According to a study by Ganle *et al*, 2014, it was revealed that most pregnant women do not trust the healthcare providers to give them quality care and it discourages some women from seeking antenatal, delivery and post-natal care. Other constraints according to the literature include verbal and physical abuse meted out to patients by health practitioners (Munguambe *et al.*, 2016; Kaba *et al.*, 2017).

CHAPTER THREE

RESEARCH METHODOLOGY

3.0 Introduction

Research often involves discovering new knowledge and integrating it with experience and current knowledge to find new patterns leading to innovative relationships. The research process is linked to the development of a framework which could contribute to the building of a new theory in a field of study or facet of reality. This section explained the way the research is carried out. It encompasses the research design, population, sampling design, data collection procedures for primary data and secondary information, data analysis, ethical considerations et.al, for the study.

3.1 Research Design

For this study, the mixed methods approach was adopted. This approach involves integration of quantitative and qualitative methods in a single study and adopts a pragmatic stance about assertions on knowledge (Creswell, 2009; Johnson & Onwuegbuzie, 2004). This approach helps build on the synergy and strength that exists between quantitative and qualitative research methods to understand our research phenomenon more fully than is possible using one method; offset the challenges associated with a single approach through triangulation (Creswell, 2009) and gave a holistic picture of the phenomenon under study (Johnson & Onwuegbuzie, 2004).

3.2 Study Area

Krowor Municipality whose capital is Nungua was carved out of the the Ledzokuku- Krowor Municipality in 2017. The municipal assembly was formed by the Local Government Act 1993(Act 462) with the legislative instrument (L.I. 2318) of 2017. It is bounded on the west by

Ledzokuku Municipalities, North by the Tema motorway, the Tema West Municipality by the eastern part and the south by the Gulf of Guinea. Females account for 51.4 percent of people living in the study area, according to forecasts from the 2010 Population and Housing Census, while the males are 82, 418 (48.6 percent) totaling 169,584. The predominant language in the municipality is Ga and Akan language.

3.3. Study Population

For this study, the target population includes mothers (lactating and other mothers) and maternal and child healthcare providers in the Krowor Municipality. It involves only females for the questionnaires and both male and females for the interview.

3.4 Sampling Technique and Sample Size

In this study, the purposive sampling technique was employed. For this study, the respondents were purposively chosen because they have specific knowledge and experiences needed for the study. Using a non-probability sampling, 96 respondents were selected from the target population (N=96) for the study. The sample for every research should be large to ensure representation (Singh, 2006). Statistically, in every research, 30 respondents (n=30) are seen as a large sample (Walliman, 2011).

3.5 Eligibility Criteria

3.5.1 Inclusion Criteria

- The respondent must be 15 years and above.
- The respondent must be a resident of Krowor Municipal Assembly for at least 4 years and must have used the healthcare facilities in the municipality.

3.5.2 Exclusion Criteria

- The respondent who does not consent to the study.
- The respondent who has not used a health facility in Krowor Municipal Assembly

3.6 Data Type, Source and Definition.

Primary data is data that is used for a specific purpose for which it was gathered (Lamb *et al.*, 2010). For this study, primary data was collected from both interviews and from the questionnaires. The use of both sources of data seeks to identify empirical and theoretical evidence with which the current study seeks to either corroborate or refute.

3.7 Data Collection Instruments

Instruments used for the data collection were questionnaire and the interview schedule (see Appendices A and B). Walliman (2011) defines an interview as a verbal interchange in which an interviewer tries to elicit information, beliefs, or opinions from an interviewee. The schedule was an unstructured one which was based on a question guide which has quite a flexible format. The interviewee has the freedom to express himself fully and it is best for in-depth studies (Walliman, 2011).

The interview was also a face-to-face one which allowed the researcher collect data from the comfort of the researched homes. This method allowed the researcher a fair idea of their socio-economic standards and their reactions and non-verbal cues to questions. These non-verbal languages were recorded to spice the information. Interviews were recorded in a book and through a recorder.

Both instruments had two main parts. The first section focused on the biographic data of respondents and the second part focuses on getting responses for the research questions. The amalgam of the two instruments was to allow for triangulation (Creswell, 2009). Here the strengths of one will offset the weakness in the other.

3.8 Validity and Reliability

Validity and reliability in research are the degree of stability exhibited when a measurement is repeated under identical conditions (Neuman, 2007). The researcher took the following steps to ensure data validity. The questionnaire and interview guide were based on information obtained from the literature review. This was to ensure that it was from a representation of elements from the subject under study. An initial draft of the research instruments was informally tested using a respondent who was not sampled for the study. Based on the feedback, few items were revised to improve better comprehension, content validity, wording format, and question flow. This will facilitate the researcher to carry out a preliminary analysis to see whether the wording and format of questions presented any difficulties when the data is analysed.

3.9 Data Processing and Analysis

For this study both qualitative and quantitative data were collected. After data was collected, the researcher organized the data. For the qualitative data, it was organized into two major categories. The first category was the locality of the respondents and the second was the gender of the respondents. After the organization, the data was then transcribed and typed into Microsoft Word. Transcription involves converting audiotapes recordings into text data. The researcher listens to the recordings repeatedly to familiarize himself with conversations and write them down. After series of readings which aimed at enabling the researcher to fully understand and become familiar with the data was done (Rossman, 2006). A thematic analysis of the data was then done. Here, the

researcher sought to generate recurring themes in the conversations based on the research questions. Questionnaires were given identifying feature in this case numbers and entered into the Microsoft Excel and the interview questions on Microsoft Word. Narratives and verbatim quotes were used to present the findings from the qualitative method analysis.

To obtain a reliable research result from the responses, Statistical Package for Social Science (SPSS vs. 20) was used for quantitative data analysis. After data was collected from the respondents, it was cross checked. The data was checked for omissions, errors, legibility, and consistency in classification. (Zikmund, 2003). Data was checked for omission and mistakes and corrections made where appropriate for analysis. Graphs and tables will be utilised in presenting the findings.

3.10 Quality Control of Data and Entry

Three research assistants recruited for the data collection exercise and were thoroughly trained all protocols and best practices in data collection processes to get accurate data devoid of biases. Pre-test was done to be familiar with the data collection instruments. The main data collection exercise took two weeks and data collected entered were strictly checked for omissions and errors. Also, cross-checked was done on the data entered to verify the accuracy and consistency.

3.11 Method of Data Analysis

The study employed descriptive statistics and interviews through thematic analysis. Descriptive statistics was such as bar charts, pie and frequencies were used to descriptive the respondents age, sex, and barriers to FMHCP. SPSS v 20 was used to compute the correlations. Also, the output from the interview was written in a form of narrative and verbatim.

Maternal mortality ratio (MMR) is given by,

$$\text{Maternal Mortality Ratio (MMR)} = \frac{\text{Maternal Death}}{\text{Total livebirths}} \times 100000 \text{ livebirths}$$

Also,

$$\text{Maternal Mortality Rate (MMR)} = \frac{\text{Maternal Death}}{\text{Population of females aged 15 – 49}} \times 1000$$

Percentage of maternal deaths among women in reproductive years is calculated by

$$\% \text{ of maternal death} = \frac{\text{Maternal Death}}{\text{Population of females aged 15 – 49}} \times 1000$$

The Pearson correlation coefficient is symbolised by the letter “r”.

The Pearson correlation formula for the coefficient r is given by:

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

Where:

r= Pearson correlation coefficient

x= Values in the first set of data

y= Values in the second set of dat

n= Total number of values

The association ranges between + 1 and -1. Positive scores mean positive relationship between x and Y, whilst negative scores mean negative relationship between X and Y, and zero scores means no relationship between X and Y.

The formula to calculate the t-score is:

$$t = r\sqrt{(n-2) / (1-r^2)}$$

where:

- **r:** The correlation coefficient
- **n:** The sample size

The p-value is calculated as the corresponding two-sided p-value for the t-distribution with n-2 degrees of freedom.

3.12 Study Variables

- Dependent Variables

Free Maternal Healthcare Policy (FMHCP)

- Independent Variables

Healthcare access

3.13 Ethical Consideration

The respondents were informed about their right to refuse to partake in the study and that their voluntary participation will help the researcher address an important educational and health issue. They were also assured that their views and contribution to the study would be treated with utmost anonymity and confidentiality. The researcher ensured the anonymity and protection of the respondents' privacy by using a coding mechanism to eliminate all identifiable details of respondents. Finally, questions regarding any challenges from the participants were addressed before the start of the survey.

CHAPTER FOUR

DATA ANALYSIS AND DISCUSSION OF RESULTS

4.1 Introduction

This chapter is focused on the study results which is followed by discussions with the relevant literature support especially the main research information. This presentation entails two main parts, the socio-demographic information of respondents and part two takes care of the core data results. The results are outlined in tables and charts, which is followed by discussion of findings. Results were displayed in charts and in some cases prose.

4.2 Data Response Rate

In all 100 questionnaires and interview questions were administered to respondents in the Krowor Municipal Assembly in this study, of which 96 were retrieved. In all, the return rate for this study was 96% which was an excellent rate for purpose of this analysis. All the questionnaires and interviews received were used for statistical analysis as there were no missing data entries or incorrect entries.

4.3 Socio-demographic characteristics of Respondents

For this study, five demographic characteristics were selected which include age, sex, marital status, educational level and religious affiliation of the respondents. This study targeted women in their reproductive ages and health practitioners in some health facilities within the Krowor municipality. Below is the summary of the descriptive statistics of the respondents.

4.3.1 Age of Respondents.

The ages of the respondents are captured in Table 1 below. From the table, 31.25% of the respondents (30 respondents) were aged between 26-35 years. 26 of the respondents representing 27.08% were also aged between 15-25 years. Of the remaining 40 respondents, 19 (19.79%) were aged 36-45, 12 (12.5%) fell within 46-55 years and 9 (9.38%) were between 56-65 years. The mean age is 34.56 years. The modal and median age fell within the age range of 26-35 years.

Table 1: Age of Respondents

Age Groups	Frequency	Percentage (%)
15- 25	26	27.08
26-35	30	31.25
36-45	19	19.79
46-55	12	12.5
56-65	9	9.38
Total	96	100

Source: Enyan-Mensah ,2022.

4.3.2 Sex of the Respondents.

For this study a major target group was women within their reproductive ages and health care providers. Figure 2 depicts the sex of the respondents. In this study 91 respondents (94.79%) were females and 5 (5.21%) were males. These five males were all health care providers within the Krowor municipality.

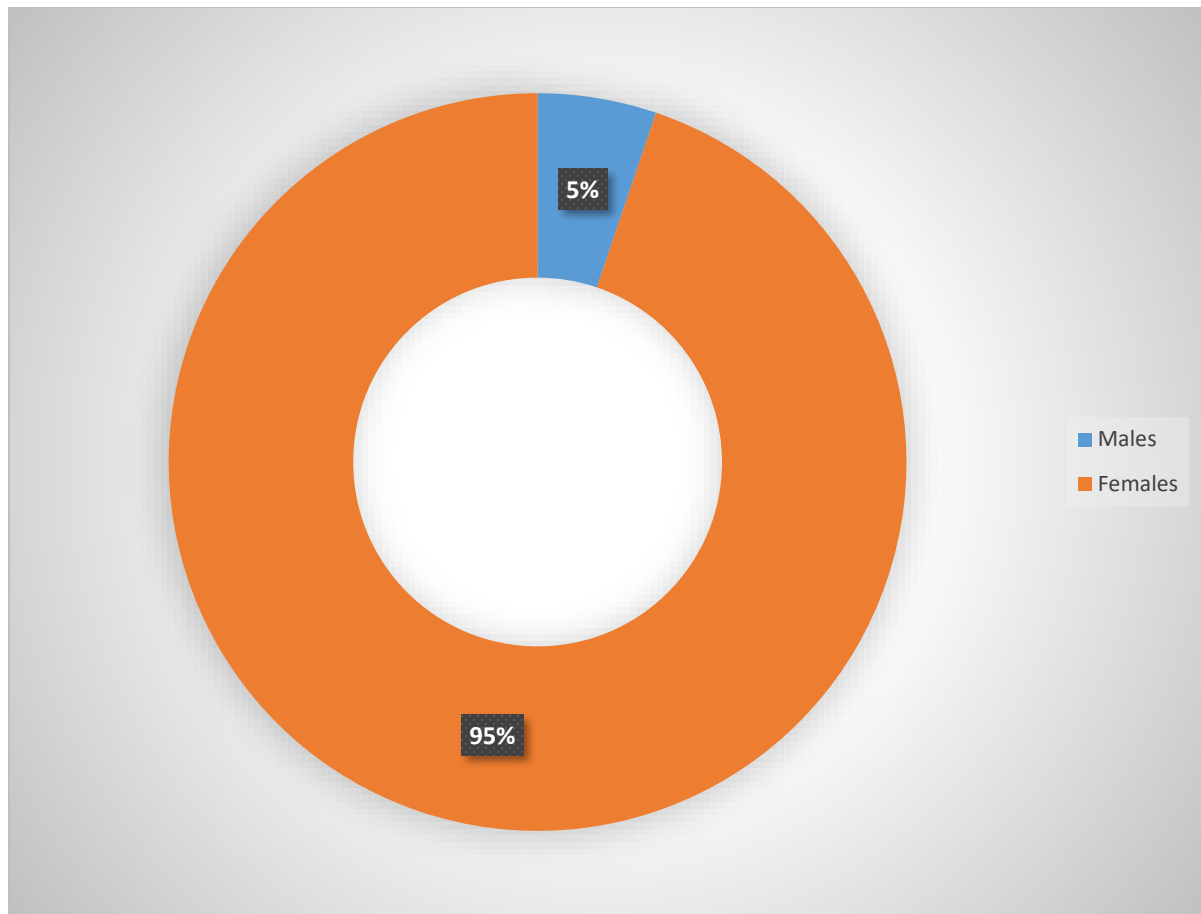


Figure 2: Sex of Respondents

Source: Enyan-Mensah,2022.

4.3.3. Marital Status

From the data for this study, 15(15.63%) of the respondents were separated/ divorced; 23 (23.95%) were cohabiting, 46 (47.92%) were married, and 12 (12.5%) were single. This is summarised in the bar graph (Figure 3) below.

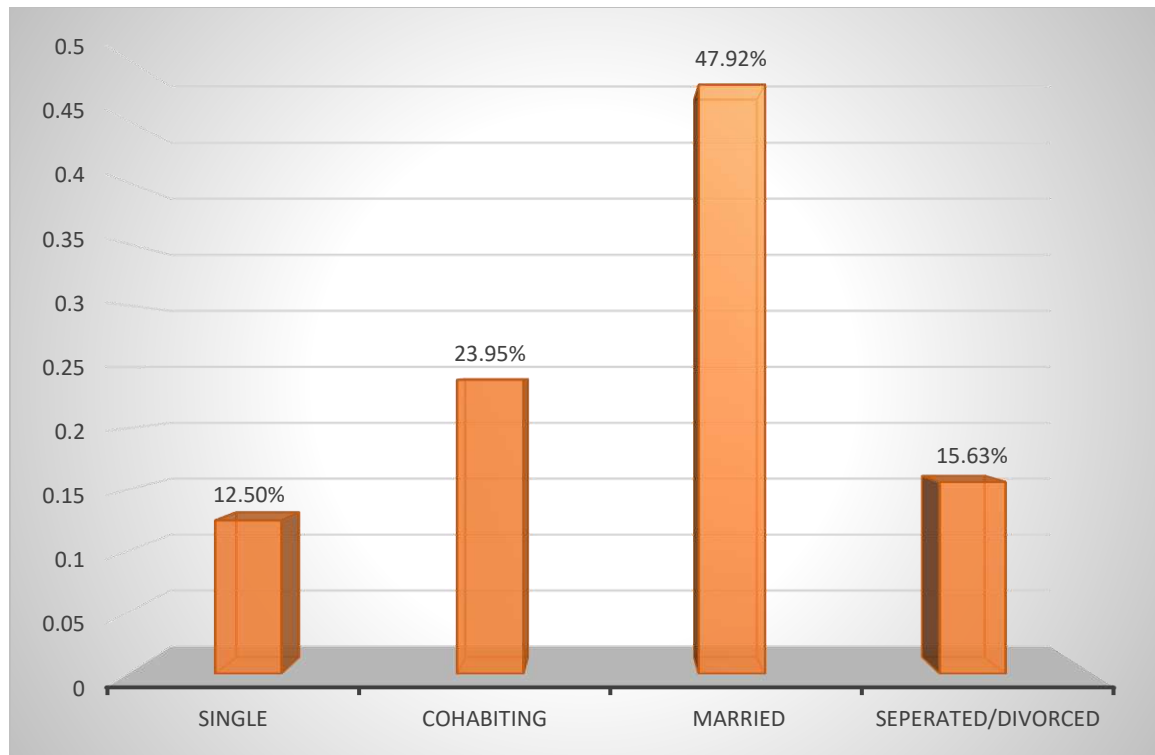


Figure 3: Marital Status of Respondents.

Source: Enyan-Mensah, 2022.

4.3.4 Educational Level of the Respondents

On the educational level, respondents were asked about their highest educational level. From the responses, 39 (40.63%) of them had senior high school or secondary education, 26 (27.08%) had tertiary education, 8 (8.3%) of the respondents are without formal education and 23 (23.96%) had basic education.

4.3.5 Religion of Respondents

Religion was an important demographic factor that was key to this study. From the responses provided 56 (58.33%) of the respondents happens to be Christians and 31 (32.29%) were Muslims.

Of the remaining 9, 2 (2.08%) were traditionalist and 7 (7.29%) were not affiliated with any religion. This is summarised in the pie chart below (Figure 4).

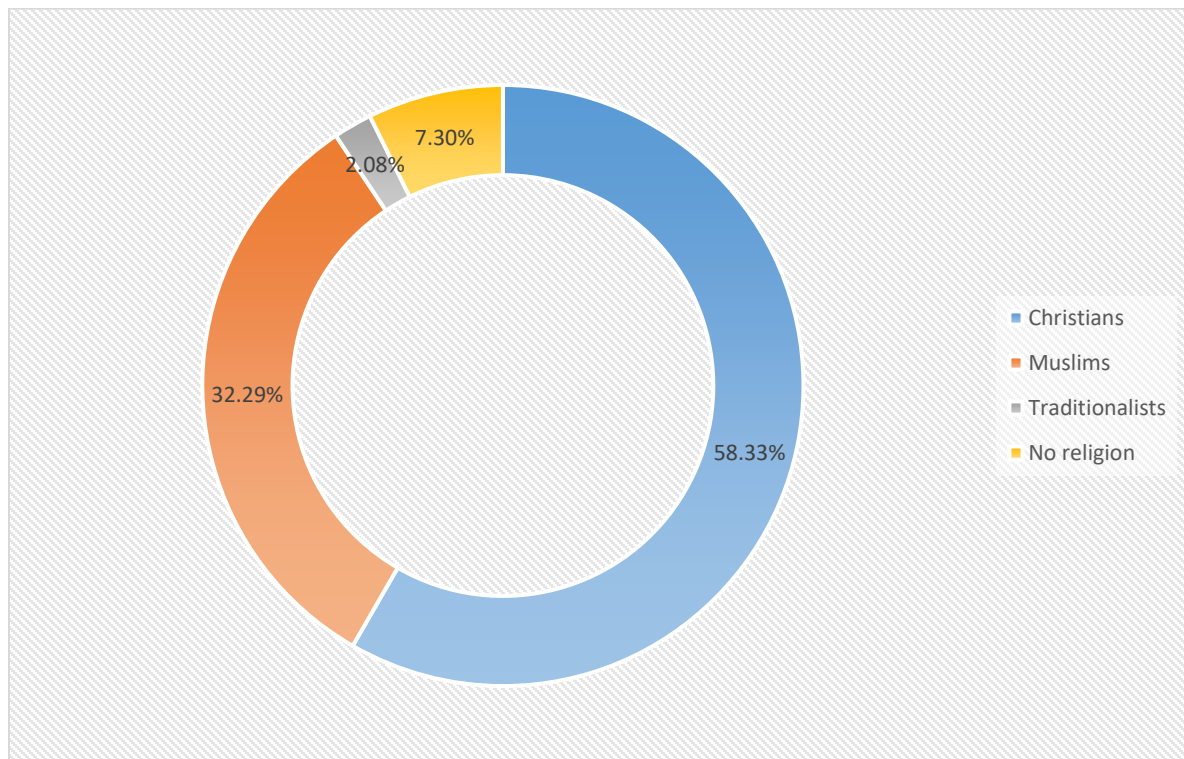


Figure 4: Religion of Respondent

Source: Enyan-Mensah,2022.

4.5. Analyses by Research Questions

The study sought to answers the questions pertaining to the impact of the FMHCP on maternal mortality and infant mortality. Also, the correlation between the FMHCP and healthcare access by expecting mothers and the challenges associated with the policy.

4.5.1 Spatial Impact of the Policy on Maternal Mortality

Based on interviews conducted with healthcare professionals within the Krowor municipality and the municipal health directorate, there are 15 health care facilities within the municipality. This is what an officer at the health directorate at the municipality had to say about the health facilities within the municipality.

“There are 15 health facilities in the municipality. Most of them are within the reach of the residents in this municipality. Of all these facilities, 13 offer free antenatal and post-natal services under the NHIS and the Free Maternal Healthcare policy. The other two are specialist hospitals. The current data on maternal death in this municipality are around 106 maternal deaths per 100,000: an improvement over data from previous years”

From the questionnaires, 88.2 % of those interviewed gave a response that they do always attend prenatal services, delivery services, and post-partum sessions at health facility within the municipality. From all the respondents, 15 had a neighbor or relative within the municipality who lost their lives due to pregnancy related complications either during or after delivery before the implementation of the FMHCP. Since the inception of the FMHCP, a reduction to 7 of such death occurred from the responses.

Additionally, a midwife in one of the facilities had this to say

“You see I have been in this field for an exceptionally long time. Before this policy, majority of the women did not have access to these services because of monetary issues. As a result of this most mothers lost their lives during pregnancy or immediately after pregnancy. This is because since they do not attend ante-natal sessions, some pregnancy related issues are not detected early and can cause their lives. However, with the ubiquitous nature of health

facilities under the FMHCP, mothers can come to the hospital and have safe delivery and post-partum checkup to know they are well.”

The findings from the study evinces a marked improvement in health outcomes of expecting mothers. There has been a notably reduction in maternal fatalities and mortalities within the municipality because of the policy. These findings corroborate that of many other authors such as Azaare *et al.*, (2020) that the policy has led to a remarkably reduction in maternal mortalities in the country.

4.5.2 Impact of the FMHCP on Infant Mortality.

The study also sought to assess the FMHCP on infant mortality within the Krowor district. The data from the responses are captured below.

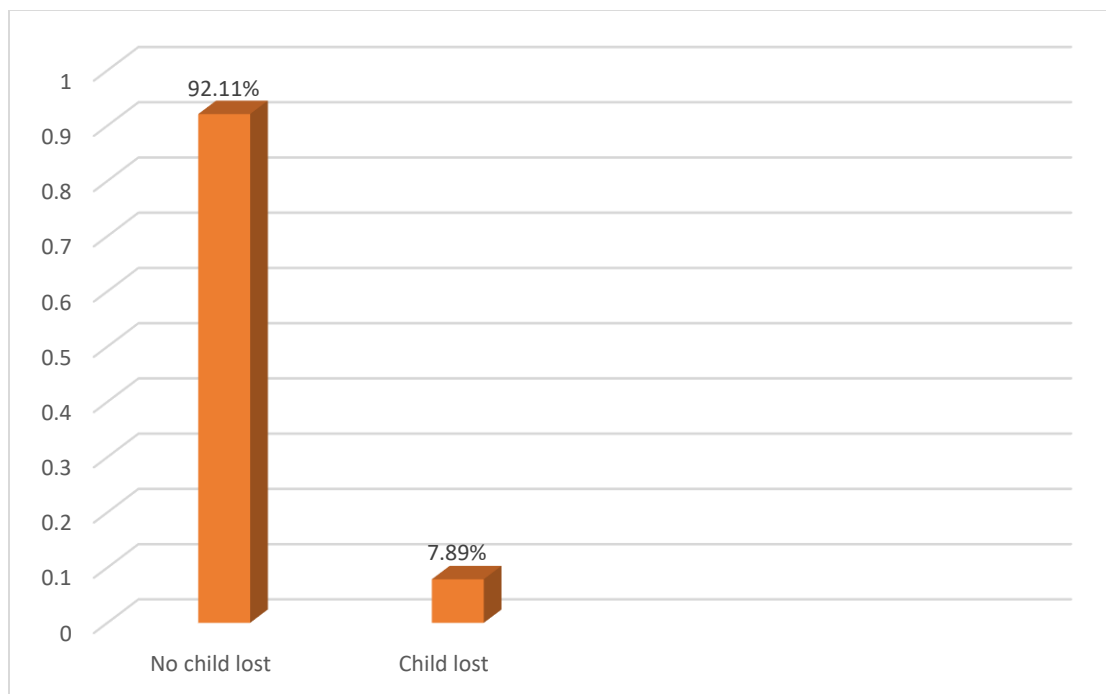


Figure 5: Infant Deaths

Source: Enyan-Mensah, 2022.

From the questionnaire results from the field, 6 (7.89%) respondents have ever lost their infants while the remaining 70 (92.11) have not lost an infant. All these infant deaths occurred before the policy was introduced. Thus, since the inception of the policy, none of the respondents have lost an infant.

From the interviews, a health practitioner reported that.

“Infant deaths in this municipality are on the decline. You see, I am a public health officer and we do well to educate the women on the need for check-ups during pregnancy. We also educate them the diet and exercise they are to engage in. This has led to a marked reduction in infant mortalities. Gone are the days where people lose their infants because they are unable to access healthcare during pregnancy. With the FMHCP, these women have access to everything they need for the sake of their infant's survival, and this has improved the life outcomes of these infants.”

Additionally, a healthcare administrator at the municipal reported a reduction on their infant mortality ratio. According to him.

“In this municipality, there has been a reduced number of infant deaths. I can say this is partly as a result of the FMHCP implementation. From our data, before the policy infant deaths were a rising toll per day pegged at around 35 a day but since the policy was introduced, there has been a marked reduction over the years. Currently, from our books, the infant deaths are less than 2 per day.”

The study reveals that the FMHCP has an impact of infant mortality within the municipality and has led to marked reduction in infant deaths in the Krowor Municipal Assembly. This finding

corroborates that of Azaare et al. (2020) and Dwomoh et al. (2020), the introduction of the FMHCP has improved delivery and care of infants and has reduced the incidence of infant mortalities.

4.5.3. Correlation between the FMHCP and Healthcare Access

The study also sought to find out the relationship between the FMHCP and healthcare access by mothers and their infants. Healthcare access was measured using the maternal healthcare services (pre-natal, delivery and post-natal services) offered by health institutions. Hence healthcare access under the FMHCP and without the policy was measured to find out how it influences expectant and lactating mothers' healthcare utilization. Also, correlation was performed using Pearson Correlation with FMHCP as a dependent variable and healthcare access as independent variables. The results are presented below:

Table 2: Healthcare Access without the FMHCP

Healthcare Access without FMHCP					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Rarely	21	27.6	27.6	27.6
	Sometimes	42	55.3	55.3	82.9
	Always	13	17.1	17.1	100.0
	Total	76	100.0	100.0	

Source: Enyan-Mensah, 2022.

Table 2 shows the questionnaire result of expectant and lactating mothers on how often they would have access healthcare services without the policy being introduced. From their responses, 21 out of 76 of them, representing 27.6 % would have rarely attended the recommended health facility visit. With 42 of the respondents, representing 55.3 % would have utilize maternal healthcare services sometimes (not always) without the policy being introduced. Finally, only 13 (17.1%) of the respondent would have always accessed the healthcare services provided for expectant and lactating mothers

Table 3: Healthcare Access under the FMHCP

Healthcare Access under FMHCP					
		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	Sometimes	9	11.8	11.8	11.8
	Always	67	88.2	88.2	100.0
	Total	76	100.0	100.0	

Source: Enyan-Mensah, 2022.

Table 3 shows the result of expectant and lactating mothers on how often they access healthcare services because of the policy. From their responses, 67 out of 76 of them, representing 88.2 % always do attend the recommended health facility visit. Only 9 respondents, representing 11.8 % do access health under the policy sometimes (not always). No respondent falls under rarely or never category. Comparing the Table 2 and Table 3 result shows a remarkably increase in the percentages or the number of expectant and lactating mothers utilizing healthcare services under the FMHCP in Krowor Municipal Assembly.

Table 4: Correlation between the FMHCP and Healthcare Access

Correlations		Healthcare Access under FMHCP	Healthcare Access without FMHCP
Healthcare Access under FMHCP	Pearson Correlation	1	.250 [*]
	Sig. (2-tailed)		.029
	N	76	76
Healthcare Access without FMHCP	Pearson Correlation	.250 [*]	1
	Sig. (2-tailed)	.029	
	N	76	76

*. Correlation is significant at the 0.05 level (2-tailed).

Source: Enyan-Mensah, 2022.

The formula to calculate the t-score is:

$$t = r\sqrt{(n-2) / (1-r^2)}$$

where:

- **r:** The correlation coefficient
- **n:** The sample size
- $t = r\sqrt{(n-2) / (1-r^2)}$
- $t = .250\sqrt{(76-2) / (1-.250^2)}$
- $t = 2.2939534045$
- **t= 2.294**

The degree of freedom (df) is =(n-2), where n: the sample size

$$\begin{aligned} df &= (76-2) \\ &= 74 \end{aligned}$$

$$\begin{aligned} \text{Percentage of Variance Explained} &= r^2 \times 100 \\ &= .250^2 (100) \\ &= 6.25 \% \end{aligned}$$

From the above table 4., the p value is 0.029

Hence the result is SIGNIFICANT at $p < 0.05$

The above Table 4 and the outputs, reveals a positive relationship between the policy and healthcare access by mothers and infants ($r=0.250$, $p<0.05$). The relationship is a low positive correlation and only 6.25 % of variance explained in healthcare access by mothers and their infants but the result is statistically significant. Based on the finding, one can argue that the FMHCP introduction has resulted in an increase in the number of pregnant women, lactating mothers and their infants who utilise healthcare in the Krowor Municipal Assembly. Pregnant women and lactating mothers can attend the recommended four antenatal care visits, delivered at health facility, and attend post-natal check-ups than before. Thus, the policy has increased the usage of maternal and infant healthcare services in the Krowor Municipal Assembly. Other factors also do influence health care access, which have been highlighted under the barriers' discussion below.

4.5.4 Barriers to the Healthcare Access under the FMHCP

The literature identified some challenges associated with the policy. The study also sought to investigate some challenges to the policy in the Krowor municipality. The information from the field indicate that the availability of facilities and health workers were not major challenges to accessing the policy. They however pointed that the long waiting hours at the facilities (45.83%) and additional costs incurred at the facilities such as laboratory services, scans, and some medications (33.33%) were the major challenges associated with the policy in the municipality. Other barriers identified were poor management (5.21%), attitude of health professionals (7.3%) and access to the facility with regards to distance (8.33%).

An expectant mother reported this.

“I attend the hospital down this road (health facility name withheld). I can go incredibly early in the morning and come back around midday. There are lots of people who use the facility. I cannot blame the workers it is just that there are many patients at this hospital. Also, you will have to pay for your scans, laboratories, and some drugs. Sometimes I wonder if it is free at all because the cost you will incur is quite high.”

Also, a woman had this to say:

“The Free maternal policy is good. It has really helped us. However, the major challenge I have is the extra cost I have to incur when I go there for either of their maternal services. They also request for certain items from you before delivery and you must have them before delivery. The most irritating part is the time you wait to be attended to. You can be in the queue for long hours and those who know people at the facility will be attended before you.”

The responses from the above shows, other costs remain a significant barrier for women seeking delivery care, postal care ad even prenatal services in healthcare institutions in the research area. This is especially crucial for poor pregnant women who do not have financial support from their families, marriages, or the men who impregnated them.

The costs which involve payments for certain services and the acquisition of certain items demanded by nurses and midwives for childbirth in the public health-care facilities. Although these costs may appear minor, they are a barrier to some women in accessing the free healthcare. This coupled with the long waiting time at these facilities are barriers to accessing the free maternal healthcare services either for an expecting mother or a mother and her infant. These findings are in line with Anafi *et al.* (2018) and Dalinjong *et al*, 2018 who reported that the long waiting hours to access maternal service and cost either direct or indirect associated with the policy serves as a challenge to most women in accessing the free maternal policy.

CHAPTER FIVE

SUMMARY OF FINDINGS, CONCLUSION AND RECOMMENDATION

5.1 Introduction

This chapter presents a summary of the main findings from the study, draws conclusions, and make recommendations for policy and future studies. The researcher was particularly interested in investigating and assessing the FMHCP on maternal and infant mortality in Krowor Municipal Assembly.

5.2 Summary of key findings.

For this study, 96 people were selected. On the respondents ages, 27.08% were between the ages of 15-25 years, 31.25% were between 26-35 years, 19.79% were aged 36-45, 12.5% fell within 46-55 years and 9.38% were between 55-65 years. The mean age is 34.56 years. The modal and median age fell within the age range of 26-35 years. Based on the nature of the study, more females were sampled than males. 94.79% were females and 5.21% were males.

On the respondents' marital statuses, 15.63% of the respondents were separated/ divorced; 23.95% were cohabiting, 47.92% were married and 12.5% were single. On the educational level, 40.63% had senior high school education, 27.08% had tertiary education, 8.3% had no formal education and 23.96% had basic education. On religion, 58.33% of the respondents were Christians and 32.29% were Muslims, 2.08% were traditionalists and 7.29% were not affiliated with any religion.

Based on the first research question, the study evinces a marked improvement in health outcomes of expecting mothers. There has been a significant reduction in maternal fatalities within the municipality because of the policy. The study also revealed that the FMHCP has an impact of

infant mortalities within the municipality and has led to marked reduction in infant deaths in the municipalities. Thus, the introduction of the free maternal healthcare policy has improved delivery and care of infants and has reduced the incidence of infant mortalities.

Based on the third research question, the study found a correlation between the free maternal healthcare policy and infants and mothers' access to health service delivery. The introduction of the policy has led to improvements in maternal and child health care service utilisation. The policy has improved the usage of maternal and infant health care services in the Krowor Municipal Assembly.

5.3 Conclusion

From the above, the following conclusions can be drawn:

The introduction of the Free Maternal Healthcare policy in Ghana and Krowor Municipality to be specific has yielded positive impacts. There are many health facilities within the district that subscribe to the governmental policy of free maternal healthcare services. This has led to accessibility of health services by mothers and resulted in a reduction in maternal mortalities and infant mortalities within the municipality.

The introduction of the policy has a correlation with maternal and infant access to health services which has led to improvements in maternal and child health care service utilisation. Mothers and pregnant women can utilize the health care services they need. The policy has improved the usage of maternal and infant health care services. The study also revealed some barriers associated with the policy which include long waiting hours at the facilities, additional costs incurred at the facilities, poor management, attitude of health professionals and access to the facility.

5.4 Recommendation

The following recommendations were made based on the findings of this study:

- There should be adequate health care providers to reduce the burden on the health system in the municipal and the long waiting time at health facilities. As 45.83% of the respondents observed long waiting hours as challenge, they encounter when utilizing maternal healthcare. Also advance appointment booking system should be implemented to manage longer time spent in these health facilities.
- The policy should be reviewed to enhance its relevance and sustainability. Since its inception in 2008, there has not been major review about it. There should be refund policy system and ambulance services should be made free at all the health facilities in the municipal and in the country in general on maternal and childcare cases.
- There should be investments in health infrastructures to enhance its utilization and capacity. My observation at the health facilities I visited during my data collection in the Krowor Municipal Assembly found out that most of the hospitals and clinics in the area lack modern health equipment and infrastructures.

5.5. Suggestion for Future Studies

This study contained not exceptionally large population. In case of future studies, it would be beneficial to use a larger number of respondents for the study. However, the requisite resources and financial support would have to be made available to assist future researchers embrace the challenge that comes with working with larger populations.

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